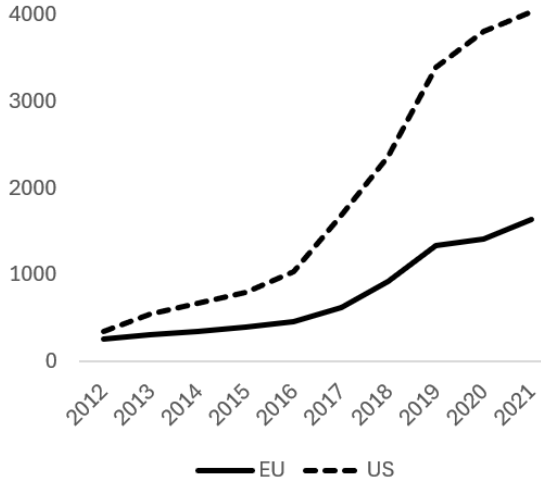
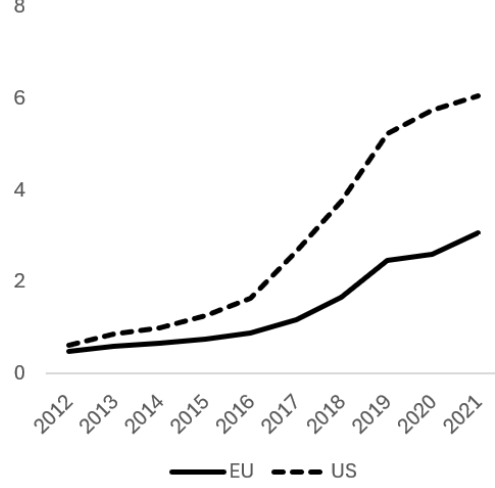


Figure 1: AI patents



(a) AI patents in domestic portfolio (count).



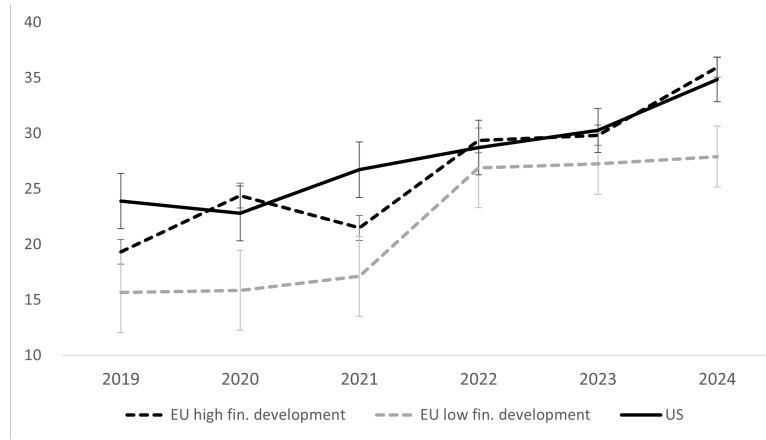
(b) Share AI patents in total domestic portfolio (%).

Note: AI patents are a subdomain of the digital patent classification based on [EPO \(2017\)](#). *Source:* PATSTAT.

The picture is somewhat more nuanced when looking at AI deployment and adoption. On average, AI adoption levels are similar in the EU and the US. Nevertheless, as shown in [Figure 2](#), the share of firms using big data and AI technology is structurally lower in EU countries with low financial development.³ There was a relatively strong catch-up in less financially developed countries in 2022 (from 17% in 2021 to 27% of firms in 2022), but a gap has opened up again in 2023-24 as adoption has remained flat. EU countries with high financial development have a similar share of firms adopting AI as in the US for the period 2019-24.

³The classification is based on the index of financial development proposed by [Betz et al. \(2025\)](#). The index uses financial market data from 2015 to 2023 and consists of two composite indicators: (i) financial market size and integration, and (ii) financial market depth. The group of EU countries with high financial development includes: Austria, Belgium, Denmark, Finland, France, Germany, Ireland, Italy, Luxembourg, Netherlands, Poland, Spain, and Sweden. The group of EU countries with low financial development includes: Bulgaria, Croatia, Cyprus, Czechia, Estonia, Greece, Hungary, Malta, Latvia, Lithuania, Portugal, Romania, Slovakia, and Slovenia.

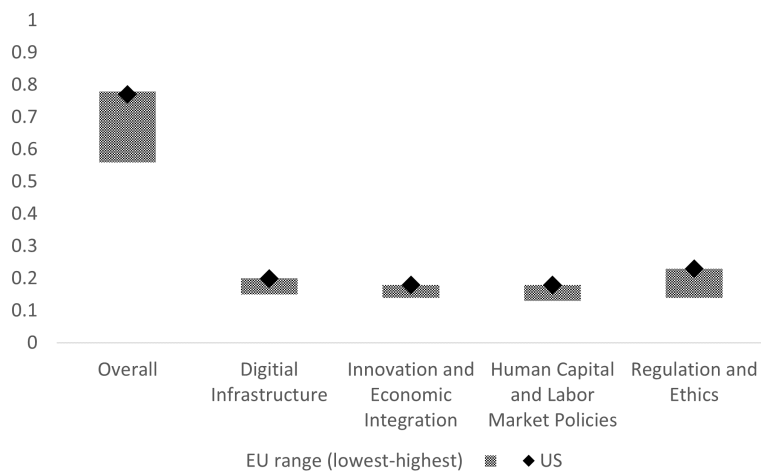
Figure 2: Use of big Data Analytics and AI (% of firms)



Note: Average share of firms reporting to use AI by country groups, controlling for firms' sector. The error bars represent 95% confidence intervals. *Source:* EIBIS 2019-2024.

Beyond EIBIS data, a benchmark indicator of AI preparedness points to the US being more conducive to AI adoption and better suited to make the most of it. Figure 3 shows the IMF's AI preparedness index, which measures how ready countries are to adopt AI based on factors like digital infrastructure, human capital, innovation, and regulation (Cazzaniga et al., 2024). Across the key categories of the index, the US scores lie in the upper range of scores for the European countries in our sample.

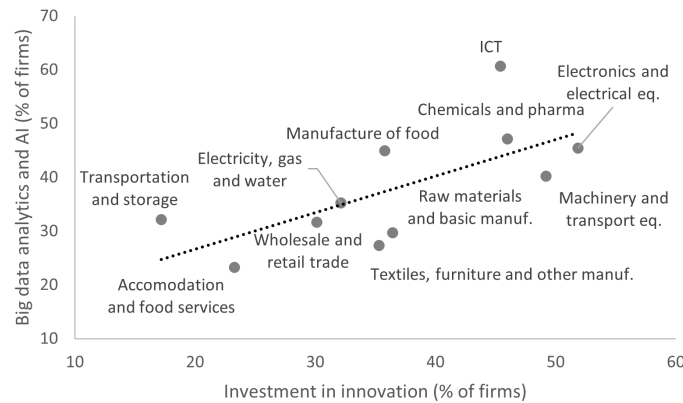
Figure 3: AI Preparedness Index



Note: The AI Preparedness Index for the EU ranges from the minimum to the maximum across EU countries. *Source:* Cazzaniga et al. (2024).

Beyond cross-country differences, AI adoption also varies significantly across sectors. This variation is driven by several structural and operational factors that influence both the applicability and the perceived benefits of AI technologies. Sectors such as Information and Communications Technology (ICT) and manufacturing tend to be early adopters due to their high data intensity, well-established digital infrastructure, and clear use cases for automation and predictive analytics. These industries often possess the technical capabilities and investment capacity required to integrate AI into core business processes, unlike a sector like accommodation and food services, where firms are significantly smaller on average. Some sectors may thus face greater barriers to adoption, including lower levels of digital maturity, limited availability of structured data, and fewer standardized AI applications. Additionally, the regulatory environment, workforce skill levels and cultural openness to technological change play a critical role in shaping sectoral adoption patterns. For example, stringent data privacy regulations can slow down implementation. The cross-sector differences are illustrated by the strong correlation between investment in innovation and AI adoption. Figure 4 shows AI uptake across EU firms, broken down by 12 sectors and corresponding innovation levels. While high-tech sectors – such as ICT and pharmaceuticals – are unsurprisingly among the highest adopters, some traditional sectors, like food manufacturing, also show above-average levels of AI uptake.

Figure 4: AI and investment in innovation by industrial sector (% of firms, 2024)



Note: Firms are weighted by value added. *Source:* EIBIS 2024.

On a more granular level, there are several firm-level factors which correlate with AI adoption. To keep the list transparent and manageable, we group EIBIS and ORBIS variables into stylised groups in terms of financial and non-financial metrics, which help

to distinguish between AI adopters and non-adopters. Table 1 reports that firms adopting AI are larger on average, invest more per employee, are more innovative and are more likely use strategic business monitoring systems. They also tend to have less problems with access to finance, since they are more likely to rely on alternative sources of financing, such as equity or bonds, they are more likely to be publicly listed, and they are less likely to be finance constrained.

Table 1: Descriptive statistics

		AI Mean	Non-AI Mean	Obs	All firms St. Dev.	p10	p90
Productivity	Labor productivity (log)	12.1	11.9	59,208	1.10	10.6	13.2
Age	Young (less than 10 years)	0.06	0.08	60,868	0.26		
Size	SMEs	0.29	0.52	60,868	0.50		
	Size (log of total assets)	17.6	16.4	47,414	2.10	13.8	19.3
Investment	Investment/employee (log)	8.80	8.30	53,058	1.70	6.20	10.5
	Fixed assets growth	0.13	0.13	46,033	0.75	-0.18	0.38
Access to finance	Newly issued bonds/equity	0.04	0.01	60,868	0.14		
	Publicly listed	0.04	0.01	60,567	0.14		
	Financial leverage (debt/total assets)	0.23	0.20	37,886	0.24	0	0.51
	Financial constraints	0.05	0.06	58,013	0.23		
Innovation	Innovator	0.49	0.37	59,448	0.49		
	Innovation new to the country	0.06	0.03	59,448	0.20		
	Innovation new to the global market	0.11	0.06	59,448	0.26		
Management	Business monitoring system	0.71	0.46	59,453	0.50		
	Profitability (EBITA/total assets)	0.13	0.11	41,863	0.15	0	0.26
	Profitability (Net income/total assets)	0.07	0.06	41,678	0.14	-0.04	0.20

Note: Labor productivity is calculated as turnover per employee. Innovators are companies which developed or introduced new products, processes or services that can be new to the company, the country or the global market. Business monitoring systems describe firms using a formal strategic business monitoring system that compares the firm's current performance against a series of strategic key performance indicators (KPIs). *Source:* EIBIS 2019-2024.

The heterogeneity in AI development and adoption between the US and EU is at the core of our empirical approach. By leveraging detailed survey and financial data, our analysis aims to capture the exogenous component of AI adoption for EU firms, using data on similar firms in the US. This approach helps mitigate endogeneity bias in the subsequent regression analysis, which may arise if AI adoption is correlated with unobserved firm-specific factors, such as managerial quality, innovation culture or strategic priorities. By using AI adoption patterns in the US – where firms operate under different institutional and policy environments – as a benchmark, we isolate variation in AI uptake